

01. The Office Gynecology Decision Model

Companion reading handout for simultaneous listening.

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The first skill in office gynecology is not choosing the treatment.

It is knowing what problem you are actually treating.

That sounds obvious until you listen to how patients enter the room.

A patient says, I am bleeding.

That sentence is not a diagnosis. It may be abnormal uterine bleeding. It may be early pregnancy loss. It may be ectopic pregnancy. It may be a medication effect. It may be endometrial hyperplasia. It may be a cavity-distorting fibroid. It may be ovulatory dysfunction. It may be cervical disease. It may be bleeding from a site that is not uterine at all.

A patient says, I have discharge.

Again, that is not a diagnosis. It may be bacterial vaginosis, trichomoniasis, candidiasis, physiologic discharge, vulvar dermatitis, genitourinary syndrome of menopause, desquamative inflammatory vaginitis, cervicitis, or a patient trying to describe odor, irritation, itching, burning, or pain with one imprecise word.

A patient says, I feel pressure.

That may be pelvic organ prolapse. It may be constipation. It may be urinary retention. It may be a fibroid uterus. It may be an adnexal mass. It may be pregnancy. It may be a pelvic floor pain syndrome. It may be something nongynecologic that happens to be felt in the pelvis.

A patient says, I need birth control.

That is also not a single question. It may be a question about pregnancy prevention, bleeding control, estrogen safety, thrombosis risk, migraine, postpartum physiology, transplant status, drug interactions, privacy, reversibility, fertility planning, or all of these at once.

So the dominant model for this whole course is simple:

Symptoms are entrances, not diagnoses.

The resident's job is to walk through the entrance without being captured by the first label.

The resident reflex is this: classify first, locate danger second, test only when the test changes management, treat in a way that fits the patient's physiology and goals, and follow until the risk is truly resolved.

Let us make that practical.

There are five repeated tasks in this course.

First, classify the problem. What category are we in?

Second, locate danger. Is this benign, dangerous, or uncertain in a way that could become dangerous?

Third, choose the least harmful test that changes management. Not the most familiar test. Not the most dramatic test. The test that actually moves the decision.

Fourth, match treatment to the mechanism, the anatomy, the reproductive goal, and the medical risk.

Fifth, follow until the condition is truly resolved.

That last step is where many good first decisions become unsafe care.

A patient treated for ectopic pregnancy is not finished when methotrexate is injected. She is finished when hCG has resolved and rupture risk has passed.

A patient with abnormal bleeding is not finished because one blind biopsy was benign if bleeding continues. A focal lesion can be missed. Persistent bleeding has to be respected.

A patient with trichomonas is not finished if the partner remains untreated and reinfection is likely.

A patient with a pessary is not finished at fitting. She needs a care plan that matches whether she can remove and clean it herself.

So this opening lecture is about disciplined first moves.

Now we will walk through the major doors of the course.

Bleeding

The wrong first question is, which pill should I give?

The right first question is, could she be pregnant?

In a reproductive-aged patient, pregnancy must be actively considered when it is clinically relevant. If she is pregnant, the pathway changes immediately. Now the issue is not generic abnormal uterine bleeding. Now the issue is location, viability, stability, and safety.

Is there an intrauterine pregnancy? Is it viable? Is she stable? Is there pain? Is there a pregnancy of unknown location? Could this be ectopic?

This matters because early pregnancy care contains one of the most important safety principles in gynecology: the diagnosis has to be safer than the treatment.

ACOG's early pregnancy loss guidance is explicit about why. Bleeding and cramping can occur in normal pregnancy, ectopic pregnancy, molar pregnancy, and early pregnancy loss. Treatment before confirmation can interrupt a normal pregnancy or cause harm. That is why ultrasound criteria for nonviability are conservative. The point is not to make residents memorize harsh numbers. The point is to prevent a false diagnosis of loss.

Ectopic pregnancy teaches the same principle from the other side. ACOG's ectopic pregnancy guidance says hCG values alone should not diagnose ectopic pregnancy. They must be interpreted with symptoms and ultrasound. A pregnancy of unknown location is not a diagnosis. It is a temporary state that requires disciplined follow-up.

Listen to the clinical tension

If the patient is unstable, uncertainty is not a reason to wait. Instability is danger.

But if the patient is stable and would continue the pregnancy if it is intrauterine and viable, uncertainty is not failure. Stable uncertainty can be managed with repeat ultrasound, serial hCG, and time.

The resident reflex is: emergencies need action; stable uncertainty needs discipline.

Suppose pregnancy has been excluded

Bleeding now becomes an AUB problem.

ACOG's abnormal uterine bleeding guidance gives the language: PALM-COEIN. PALM is structural: polyp, adenomyosis, leiomyoma, malignancy and hyperplasia. COEIN is nonstructural: coagulopathy, ovulatory dysfunction, endometrial, iatrogenic, and not yet classified.

Do not hear PALM-COEIN as a list for an exam. Hear it as protection against lazy diagnosis.

Heavy bleeding is not one disease.

Imagine a teenager with heavy bleeding since menarche. The dangerous branch may be a bleeding disorder. ACOG notes that up to twenty percent of women presenting with heavy menstrual bleeding may have an underlying bleeding disorder. That is not a detail to decorate the history. It tells you why you ask about postpartum

hemorrhage, surgical bleeding, dental bleeding, bruising, epistaxis, gum bleeding, and family history.

Now imagine a forty eight year old with obesity and persistent bleeding. The dangerous branch is different. Endometrial hyperplasia or malignancy must be considered. ACOG recommends endometrial sampling as a first-line test in patients with AUB older than forty five, and also in younger patients with unopposed estrogen exposure, failed medical management, or persistent AUB.

Now imagine a patient with heavy bleeding and a cavity-distorting submucosal fibroid. The problem is not simply "she bleeds." The problem is structural anatomy. A levonorgestrel IUD may be excellent for many bleeding patients, but if the cavity is significantly distorted, placement and retention may fail. A medication can quiet the endometrium, but it cannot make a distorted cavity normal.

Same complaint. Three different problems.

Bleeding is the entrance. Classification chooses the hallway.

Move to contraception

Contraception is often taught as a menu. That is too shallow for residents.

The FDA Birth Control Guide gives one important axis: typical-use effectiveness. Sterilization, IUD or IUS methods, and the implant are in the less-than-one pregnancy per one hundred women in the first year group. They are highly effective partly because they remove repeated human behavior from the system. Pills, patch, ring, condoms, diaphragm, sponge, cervical cap, internal condom, spermicides, and app-based methods depend more on repeated correct use.

That is a useful ladder.

But it is not the whole decision.

The FDA chart tells you how methods compare in broad actual use. ACOG's hormonal contraception guidance and the US MEC tell you who can safely use which method.

Those are different jobs.

This distinction prevents many errors.

A patient asks for "the best contraception." If you answer with a product name, you have answered too early.

Best at preventing pregnancy? Best for heavy bleeding? Best without estrogen? Best without a pelvic procedure? Best for someone taking enzyme-inducing medication? Best for someone with migraine with aura? Best for someone after renal transplant with graft failure? Best for someone who wants pregnancy next year?

ACOG teaches that the US MEC categories are risk balancing against unintended pregnancy, not risk compared with nothing. Category one means no restriction. Category two means advantages generally outweigh risks. Category three means risks usually outweigh advantages, though the method may be used when better options are unacceptable or unavailable. Category four means unacceptable health risk.

That framework matters because the same method can be right in one body and wrong in another.

Estrogen-containing contraception may be acceptable for one patient and dangerous for another. A twenty five year old smoker is not the same as a smoker older than thirty five. MTHFR mutation is not the same as antiphospholipid syndrome. Migraine without aura is not migraine with aura. A stable solid-organ transplant is not graft failure. A family history of breast cancer is not current breast cancer.

The resident reflex is: name the method, name the risk mechanism, then name the safer alternative.

Move to vaginal symptoms

The common error is treating discharge by appearance alone.

The better model is ecosystem, organism, and inflammation.

ACOG's vaginitis guidance starts with vaginal biology. Estrogen increases glycogen in vaginal epithelial cells. Glycogen supports lactobacilli. Lactobacilli produce lactic acid. Lactic acid keeps the reproductive-age vaginal pH usually below four point five. That acidic ecosystem helps suppress pathogenic overgrowth.

Once you understand that, the office tests stop feeling arbitrary.

pH is not a strip of trivia. It is a readout of the ecosystem.

Microscopy is not a ritual. It asks what cells and organisms are actually present.

The whiff test is not folklore. It detects amines in a dysbiotic anaerobic environment.

Bacterial vaginosis is dysbiosis. Lactobacilli decrease, anaerobes increase, pH rises, clue cells appear, and the discharge may be thin and fishy. It is associated with sexual activity, but it is not simply one pathogen transmitted from one partner.

Trichomonas is different. It is a sexually transmitted protozoan infection. Partner management matters. NAAT is preferred because microscopy misses too many cases.

Candidiasis is different again. It is yeast-associated inflammation. pH is usually normal. Symptoms alone are not enough because Candida can colonize without causing disease, and noninfectious vulvar disease can mimic yeast.

Now imagine a patient who has treated herself for yeast five times and still has burning.

The resident who sees discharge and prescribes fluconazole again has not diagnosed the problem. The resident who asks, where are the symptoms, what is the pH, what does microscopy show, is there objective yeast, is there vulvar dermatitis, is this genitourinary syndrome of menopause, has changed the quality of care.

Vaginitis is not "which discharge looks like which organism." It is ecosystem plus organism plus inflammation.

Move to pelvic organ prolapse

Prolapse teaches one of the most important principles in all of office gynecology: anatomy alone is not disease.

ACOG's prolapse guidance states that mild descent is common and should not automatically be considered pathologic. Pelvic organ prolapse is a problem when it causes pressure or bulge symptoms, sexual dysfunction, lower urinary tract dysfunction, or bowel dysfunction.

That means the patient with stage two descent and no bother is not a failed diagnosis. Reassurance can be correct treatment.

And the patient with bothersome bulge, splinting, incomplete emptying, and sexual dysfunction deserves more than, "It is benign."

POP-Q gives the shared map. It documents the anterior wall, apex, posterior wall, genital hiatus, perineal body, and total vaginal length. It matters before treatment because it makes anatomy reproducible.

But do not treat the POP-Q number in isolation.

Treat symptoms and function.

The source gives a useful anatomic clue: many women feel symptoms when the leading edge reaches about zero point five centimeters beyond the hymenal ring. That number is not magic. It reminds you that the hymen is a clinically meaningful reference point because symptoms often appear when descent reaches the outlet.

So the resident reflex is: map the compartments, identify the apex, ask what function is impaired, and do not confuse measurement with indication.

Move to adnexal masses

An adnexal mass is another place where the first move matters.

The goal is not to make every cyst disappear. The goal is to exclude malignancy while avoiding unnecessary surgery for benign disease.

ACOG's adnexal mass guidance teaches that transvaginal ultrasound is the recommended initial imaging modality for a suspected or incidentally identified pelvic

mass. CT, MRI, and PET are not the initial characterization tests.

That sentence deserves to be understood, not memorized.

Ultrasound gives morphology: cystic, solid, mixed, septations, mural nodules, papillary projections, free fluid, laterality, Doppler flow. It lets you describe the object.

CT becomes useful when cancer is suspected and the question changes from, what is this mass, to, is there metastatic disease? Ascites, omental disease, peritoneal implants, nodes, liver lesions, obstructive uropathy, or another primary cancer.

So ordering CT first for a simple initial adnexal mass is not more sophisticated. It may be the wrong tool for the first question.

Listen to the risk language

Simple thin-walled cyst, smooth walls, no solid component, no septations, no internal blood flow: reassuring.

Size greater than ten centimeters, papillary or solid components, irregularity, ascites, high color Doppler flow: concerning.

CA 125 is also context-dependent. It means more after menopause. In premenopausal women it can rise with endometriosis, pregnancy, pelvic inflammatory disease, fibroids, inflammatory disease, and other conditions. A postmenopausal woman with a pelvic mass and elevated CA 125 is a different problem from a premenopausal woman with a suspected endometrioma and a moderate elevation.

The resident reflex is: ultrasound morphology first, menopausal context second, markers as context, not as a universal cancer detector.

Now return to early pregnancy, because it is the clearest example of how this course thinks.

Early pregnancy care often asks the resident to tolerate uncertainty without becoming passive.

If the patient is unstable, you act.

If the patient is stable, you may need to wait carefully.

That is a hard discipline because residents often feel that every uncertain result demands immediate closure. But premature closure is dangerous.

In early pregnancy loss, the danger is diagnosing nonviability too early and interrupting a viable pregnancy.

In pregnancy of unknown location, the danger is treating with methotrexate before intrauterine pregnancy has been reasonably excluded.

In ectopic pregnancy, the opposite danger is false reassurance. Falling hCG does not completely exclude ectopic rupture. Low hCG does not make rupture impossible. The follow-up is not administrative. It is safety.

So the model is not "wait" versus "treat." The model is stable uncertainty versus unstable danger.

Stable uncertainty gets structure: repeat ultrasound, serial hCG, clear return precautions, documented follow-up, and patient values.

Unstable danger gets urgent treatment.

Now let us rehearse the course model with a few short cases.

Case one.

A forty eight year old with obesity has persistent irregular bleeding. Pregnancy test is negative. A blind biopsy returns benign proliferative endometrium, but bleeding continues.

The tempting error is reassurance. The better move is to remember that a blind biopsy is stronger when it finds disease than when it excludes focal disease. Persistent bleeding after benign pathology requires further evaluation for focal or structural pathology, such as polyp or leiomyoma, with imaging, sonohysterography, or hysteroscopy as appropriate.

The lesson: follow until the clinical question is actually resolved.

Case two.

A patient wants contraception. She has migraine with aura. The FDA chart shows combined pills, patch, and ring as reasonably effective typical-use methods.

The tempting error is to stay on the effectiveness ladder. The better move is to switch from effectiveness to eligibility. Migraine with aura changes estrogen safety. The resident should offer safer effective alternatives, such as progestin-only or nonhormonal methods depending on the rest of the history.

The lesson: efficacy starts the conversation; medical risk chooses the method.

Case three.

A patient has discharge and odor after intercourse. The pH is elevated, microscopy shows clue cells, and whiff is positive.

That is a dysbiosis pattern. It is not a reason to culture the vagina for BV. It is not a reason to treat the partner routinely. It is a reason to treat symptomatic bacterial vaginosis and counsel with the correct biology.

Now change the case. NAAT confirms trichomonas. Now partner treatment matters because the biology is different.

The lesson: similar symptoms do not mean similar transmission or similar follow-up.

Case four.

A patient has stage two prolapse on examination but no bulge, no pressure, no voiding dysfunction, no bowel dysfunction, and no sexual dysfunction.

The tempting error is to treat the measurement. The better move is education and reassurance.

Now change the case. Same stage, but the patient splints to defecate, avoids exercise because of bulge, and has incomplete emptying. Now the same anatomy has become clinically meaningful.

The lesson: the patient tells you whether the map matters.

Case five.

A stable patient has pelvic pain, a positive pregnancy test, no definitive intrauterine pregnancy on ultrasound, and she would continue the pregnancy if it is viable.

The tempting error is methotrexate for uncertainty. The better move is disciplined follow-up unless clinical danger changes the pathway. Repeat ultrasound, serial hCG, and careful counseling protect the patient from both ectopic rupture and unnecessary exposure of a viable intrauterine pregnancy to methotrexate.

The lesson: stable uncertainty is a management state, not a failure.

Pull the thread together

This course is not a collection of unrelated office topics.

It is one repeated clinical habit applied to different problems.

Bleeding becomes pregnancy status, PALM-COEIN, endometrial risk, structural anatomy, and hemostasis.

Contraception becomes effectiveness, user dependence, hormone exposure, medical eligibility, and reproductive goals.

Vaginitis becomes ecosystem, organism, inflammation, and partner logic.

Prolapse becomes symptoms, function, compartments, and patient goals.

Adnexal masses become patient context, ultrasound morphology, menopausal status, markers, and referral threshold.

Early pregnancy becomes stability, location, viability, diagnostic certainty, and follow-up to resolution.

The details in this course matter. The numbers matter. The names matter. PALM-COEIN, POP-Q, US MEC, CA 125, discriminatory hCG, Amsel criteria, methotrexate follow-up, and ultrasound morphology all matter.

But they should never feel like disconnected facts.

Each one is a safety tool.

PALM-COEIN prevents you from calling every bleeding patient hormonal.

US MEC prevents you from treating contraception as a menu without physiology.

pH and microscopy prevent you from treating discharge by appearance.

POP-Q prevents vague prolapse documentation, but symptoms prevent overtreatment.

Ultrasound morphology prevents both cancer delay and unnecessary surgery.

Conservative early pregnancy criteria prevent harm to viable pregnancies.

Serial hCG follow-up prevents false closure after ectopic treatment.

The final resident reflex for this lecture is this:

Do not rush to the treatment line.

Classify first.

Locate danger second.

Choose the test that changes management.

Treat the mechanism, the anatomy, the medical risk, and the patient's goal.

Then follow until the risk is truly gone.

If you hold that model, the rest of the course becomes much easier. Not easier because the facts are simple. Easier because every fact now has a job.

That is the foundation.